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标题	Beyond Wittig Olefination: Phosphorus Ylide as a Ring-Expansion Reagent for Dibenzocycloheptanone Synthesis
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摘要（原文节选）

We report an unprecedented transformation between phenanthrene-9,10-diones and phosphorus ylide, where methylenetriphenylphosphane exhibits dual reactivity patterns, enabling both olefination and ring expansion to construct seven-membered carbocycles in moderate to good yields. This process proceeds through initial Wittig olefination followed by an unusual alkenyl group 1,2-migration pathway from either an oxaphosphetane or betaine intermediate. Mechanistic studies, including deuterium-labeling experiments and isol ⋯

数据表 · 1（11 行，含数值行 11 行）

Reaction Condition Optimization

entry	wittig reagent (x equiv)	solvent	yield/%
1	10.0	THF	77%
2	8.0	THF	63%
3	5.0	THF	80%
4	3.0	THF	22%
5	1.0	THF	NR
6	5.0	1,4-Dioxane	38%
7	5.0	Toluene	ND
8	5.0	CH2Cl2	ND
9	5.0	DMSO	ND
10	5.0	DMF	ND
11	10.0	THF	ND

脚注（逐字）：aReaction conditions: The Wittig reagent was prepared in situ by the addition of n-BuLi (2.5 M in hexanes, x equiv) to a mixture of CH3PPh3Br (x equiv) in THF (0.25 M) at 0 ° C for 30 min. Subsequently, 1a (0.10 mmol) and solvent (1 mL) were added to the above Wittig reagent at 0 ° C and stirred at 40

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